

Catedral 1330 — natural gas validation report

Building: Catedral 1330, downtown Santiago, Chile **Type:** Apartment Block (AB), hormigon (concrete), ~2005
Archetype: `CL.AB.intN.hor` — 2000-2015 normative period, horizontal (flat roof) **GeoNode record:**
`merlin_rcp.edificios.edificio_id = 2618695` **Script:** `santiago_ab_calibration.py`

Simulation pipeline

Step	Source	Value
Building properties	<code>merlin_rcp.edificios</code>	248 inmuebles, 9,844 m², avg 39.7 m²/unit
Census district	<code>boundaries.src0125__ine__censo2024__poly__distrital</code>	1310102 — Santiago / Moneda
Typical Meteorological Year	<code>tmy.src0101__minenergia__tmy_hourly</code>	MinEnergía national dataset, 8760 h
Archetype	<code>tsib/data/episcope/CL_episcope.csv</code>	U_Wall 1.9, U_Roof 1.5, U_Floor 0.8, U_Win 3.5 W/m²K
ACS calibration	Summer billing baseline (Jan–Mar 2025)	35.3 L/person/day (heating = 0 in summer)
Simulation engine	<code>tsib.Building5R1C.sim_demand_direct()</code>	ISO 13790 5R1C, forward-Euler, no LP solver
Scale-up	219 residential units	(248 DB units includes commercial/parking)

TMY profile — district 1310102 (Santiago / Moneda)

Month	Mean T (°C)	Mean GHI (W/m²)
Jan	21.9	358
Feb	22.1	321
Mar	19.1	256
Apr	16.6	187
May	11.1	107
Jun	10.6	91
Jul	9.8	88
Aug	12.0	126
Sep	13.6	193
Oct	17.2	257
Nov	18.9	326
Dec	20.7	366

Annual summary

Item	Value
Archetype ref area (CSV)	58 m ²
Heating demand (simulated)	49 kWh/m ² /a (archetype unit)
Annual heating — whole building	623 MWh/a
Annual ACS (calibrated)	369 MWh/a
Gas — heating	67,815 m ³ /a
Gas — ACS	40,182 m ³ /a
Gas — total simulated	107,997 m ³ /a
Gas — billed (Jul 2024 – Jun 2025)	106,292 m ³
Annual error	+1.6%

Monthly comparison

Month	Billed (m ³)	Sim-heating (m ³)	Sim-ACS (m ³)	Sim-total (m ³)	Error (%)
Jul-24	19,571	17,732	3,413	21,145	+8.0
Aug-24	19,161	11,456	3,413	14,868	-22.4
Sep-24	15,400	6,092	3,303	9,395	-39.0
Oct-24	4,260	1,362	3,413	4,775	+12.1
Nov-24	4,450	156	3,303	3,459	-22.3
Dec-24	3,790	0	3,413	3,413	-10.0
Jan-25	3,520	0	3,413	3,413	-3.0
Feb-25	3,120	0	3,082	3,082	-1.2
Mar-25	3,400	570	3,413	3,982	+17.1
Apr-25	4,570	2,031	3,303	5,333	+16.7
May-25	10,750	13,434	3,413	16,847	+56.7
Jun-25	14,300	14,982	3,303	18,285	+27.9
Jul-25	15,100	17,732	3,413	21,145	+40.0

Validation metrics

Approach	Bill total (m ³)	Sim total (m ³)	Annual bias	MAE (m ³ /month)	MAPE (%)
tsib-fcr + real TMY (this work)	106,292	107,997	+1.6%	2,413	21.3
approach1 (prev. engineering, Madrid proxy)	121,392	92,308	-24.0%	2,554	22.2
approach2 (UA postcalibrated, Madrid proxy)	121,392	278,908	+129.8%	12,896	106.4

Key findings

Annual total: Model matches bills within 1.6% using a fully automated pipeline (no manual calibration beyond the ACS summer-baseline adjustment).

Monthly shape error (MAPE 21.3%): The TMY represents a *typical* year, not calendar 2024-2025. The real winter was shifted — August 2024 was the coldest billing month (19,161 m³) while the typical year peaks in July. Cold snaps and inter-annual variability are not captured by a TMY.

ACS calibration: Summer baseline (Jan–Mar 2025, when heating = 0) implies 35.3 L/person/day, lower than the commonly assumed 50 L/day. This reflects actual measured consumption at this building rather than a design value.

Underestimate in Sep 2024 (–39%): September had unusually high gas consumption suggesting a cold front not present in the typical year. This is a known limitation of TMY-based simulations.

Files

File	Description
santiago_ab_calibration.py	Full pipeline script (DB → TMY → sim → comparison)
natural_gas_consumption.csv	Raw monthly gas bills Jul-2024 to Jul-2025
tmy_santiago_1310102.csv	Real TMY extracted from GeoNode (cached)
tsib_validation_monthly_comparison.csv	Previous approach monthly data
tsib_validation_report.md	This report